

Technology Initiatives for E-Cell Operations

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Overview

The issues are not about effort; the teams work incredibly hard. The gap is in the systems supporting them. Three areas stood out to me as practical and high-impact opportunities where technology can step in and make a real difference.

1. Smart Hospitality & Room Allocation System

The Problem

Right now, the Hospitality team manages room allocation almost entirely through Excel sheets. During an event the size of E-Summit, this creates a cascade of small but costly problems. Rooms get double-assigned or incorrectly allocated. There is no live view of which rooms are occupied. Gender-separated accommodation rules are hard to enforce manually. Last-minute changes — someone does not show up, or a waitlisted group suddenly needs three more rooms — turn into a scramble. On top of all this, the help-desk volunteers have almost no visibility into where anyone is staying, which makes answering participant questions painfully slow.

The Solution

I would build a dedicated web application for the Hospi team — one system that handles everything from auto-allocation to real-time status tracking. The key capabilities would include:

- Automated room assignment based on predefined rules including gender separation and room capacity
- A live dashboard showing current occupancy across all rooms, updated instantly
- Check-in and check-out integration to keep the data accurate throughout the event
- A fast participant search tool so help-desk volunteers can answer queries in seconds
- QR code scanning at accommodation desks to speed up the check-in process

How I Would Build It

The frontend would be built in React, with Django handling the backend logic and PostgreSQL as the database. Before the event, the system would be stress-tested using simulated participant data to make sure edge cases are handled properly. It would be deployed on a cloud server with role-based access control so only authorised team members can log in.

What It Could Grow Into

Beyond E-Summit, this platform could expand to cover all E-Cell events. Transport coordination, guest itinerary management, automated check-in flows, and integration with registration databases are all natural next steps — turning a single-event tool into a full hospitality operations system.

2. AI Chatbot for E-Cell Event Websites

The Problem

Every year, the organising team fields the same questions from hundreds of participants: Where do I go? How do I register? What documents do I need? These questions are not unreasonable — but they consume enormous amounts of team time that could be spent elsewhere. The information is usually on the website, but spread across multiple pages in a way that is not easy to navigate quickly.

The Solution

I would integrate an AI-powered chatbot directly into E-Cell event websites, acting as an always-on virtual help desk. It would handle questions on:

- Event registration steps, deadlines, and confirmation
- Session schedules and speaker details
- Venue directions and campus navigation
- Accommodation details and check-in instructions
- Contact information for key team members
- Frequently asked questions drawn from previous events

Where appropriate, it would also link users directly to the relevant section of the website rather than just giving a text answer.

How I Would Build It

The first step is building a solid knowledge base — pulling from past help desk logs, social media queries, and registration FAQs. The chatbot would then be developed using modern AI APIs and natural language processing, trained on this data along with current event content. Once tested, it gets embedded as a widget across all E-Cell websites. After launch, real user queries would be monitored to continuously refine its accuracy.

What It Could Grow Into

The chatbot could eventually evolve into a broader AI assistant for E-Cell — one that guides users through registration flows, recommends events based on their interests, delivers

personalised schedules, and autonomously handles a significant share of participant support end to end.

3. AI-Animated Registration Walkthrough Videos

The Problem

Registration is the first touchpoint most participants have with E-Cell, and it does not always go smoothly. Multi-step forms confuse first-time users. Required fields get missed. Documents are not attached correctly. The result is a flood of support queries in the days before an event and a noticeable number of incomplete or incorrect registrations — extra work for both participants and the team.

The Solution

I would create a series of short, AI-generated animated videos that walk participants through the entire registration process, step by step. Each video would cover a specific part of the flow:

- Creating and verifying an account
- Browsing available events and selecting the right ones
- Filling out the registration form correctly
- Processing payment where required
- Confirming and verifying successful registration

Visual walkthroughs are far more accessible than written instructions — especially for first-time users or those registering through a mobile device.

How I Would Build It

The process begins with mapping out every discrete step of the registration flow. From there, animation tools and AI video generation platforms would be used to produce clean, professional tutorial videos. These would be embedded directly on the relevant registration pages and also shared via social media channels and confirmation emails to reach as many participants as possible.

What It Could Grow Into

This initiative could eventually become a full interactive onboarding experience for all E-Cell platforms — incorporating smart form assistance, guided product tours, and AI-directed registration flows that adapt based on what a user is struggling with.